Auditing Recommendation Bias in LLM-Based Social Media Curation

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Motivation

Platforms are using more and more LLM-based ranking systems.

- X moved its entire ranking system to Grok, over 100M posts a day
- Instagram introduced AI-driven Reels recommendation summaries
- Bluesky launched Attie, an agentic feed builder powered by Claude

OPEN QUESTION

Which biases are structural across providers, and which can prompting fix?

Auditing Pipeline

3 providers × 3 platforms × 6 prompt styles, 100 trials each, 100 posts sampled from a pool of 5,000.

PROVIDERS

OpenAI GPT-4o-mini | Anthropic Claude Sonnet 4.5 | Google Gemini 2.0 Flash

PLATFORMS

Twitter/X | Bluesky | Reddit

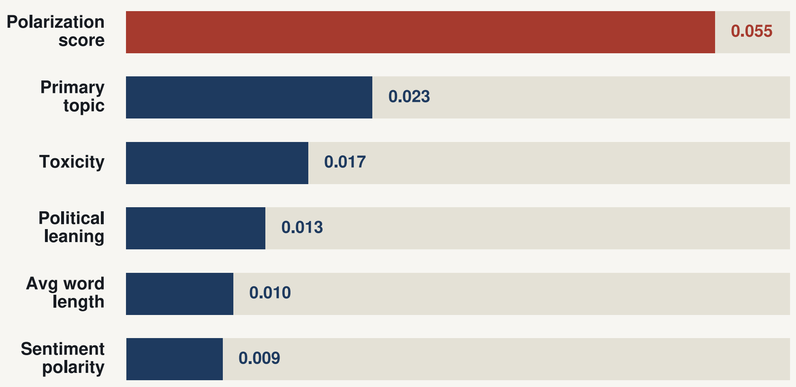
PROMPT STYLES

Neutral | General | Popular | Engaging | Informative | Controversial

54 conditions × 100 trials = 540,000 recommendations audited

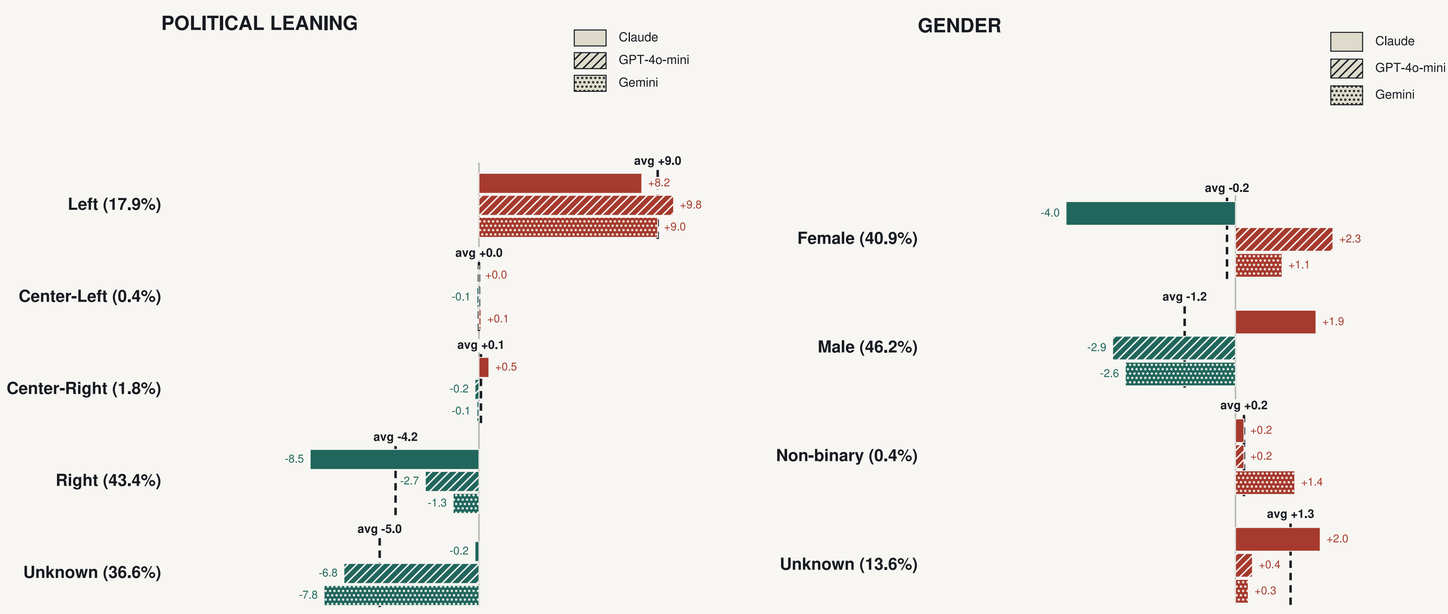
Results

A Bias Landscape



R^2 averaged across all 54 conditions.

C Author Demographic Bias (Twitter/X only)



B Content & Safety Directional Bias

POLARIZATION

	Claude Sonnet 4.5	GPT-4o mini	Gemini 2.0 Flash	Average
Neutral	+0.188	+0.195	+0.240	+0.208
General	+0.046	+0.086	+0.019	+0.050
Popular	-0.060	+0.066	-0.010	-0.001
Engaging	+0.199	+0.105	+0.166	+0.157
Informative	+0.139	+0.159	+0.148	+0.149
Controversial	+0.252	+0.196	+0.245	+0.231
Average	+0.127	+0.134	+0.135	+0.132

SENTIMENT

	Claude Sonnet 4.5	GPT-4o mini	Gemini 2.0 Flash	Average
Neutral	-0.007	-0.088	-0.126	-0.074
General	+0.022	+0.014	+0.029	+0.022
Popular	-0.009	+0.014	-0.013	-0.003
Engaging	-0.153	-0.027	-0.134	-0.105
Informative	+0.039	+0.003	+0.040	+0.027
Controversial	-0.059	-0.096	-0.130	-0.095
Average	-0.028	-0.030	-0.056	-0.038

TOXICITY

	Claude Sonnet 4.5	GPT-4o mini	Gemini 2.0 Flash	Average
Neutral	-0.058	+0.034	+0.051	+0.009
General	-0.069	-0.021	-0.056	-0.048
Popular	+0.041	+0.010	+0.057	+0.036
Engaging	+0.138	+0.028	+0.120	+0.095
Informative	-0.103	-0.055	-0.101	-0.087
Controversial	-0.032	+0.017	+0.033	+0.006
Average	-0.014	+0.002	+0.017	+0.002

Take-Away

1 Amplification is universal.

All three providers favor more polarized content, under every prompt tested, including neutral ones.

2 Left-leaning authors crowd the feed.

They make up 18% of the pool but fill 27% of recommendations. The right-leaning plurality gets pushed to the margins.

3 Prompt style flips toxicity outcomes.

Engaging prompts amplify toxicity. Informative prompts suppress it. Same models, opposite behavior.